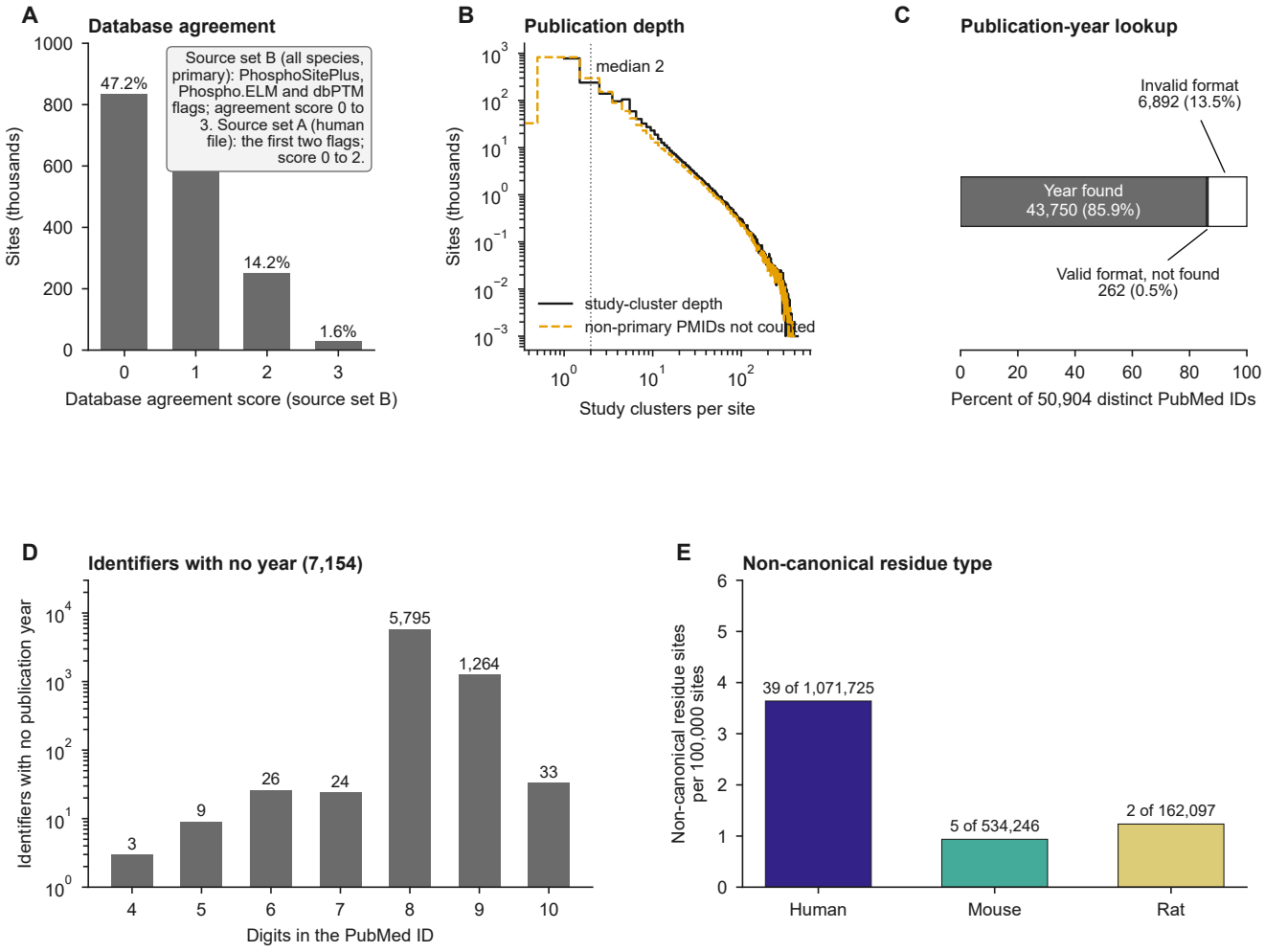
Supporting Information

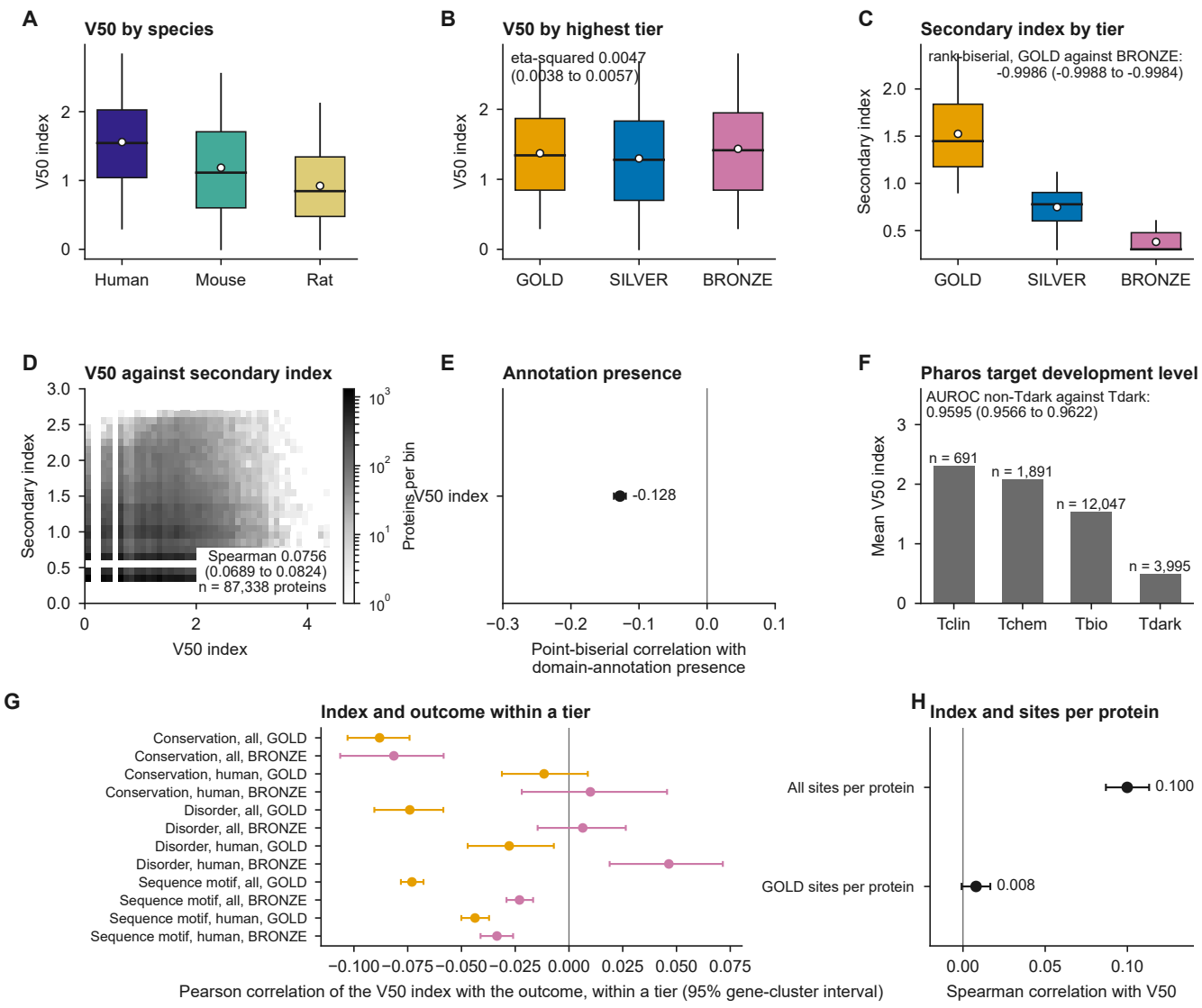
Table S1	Contrast grid (534 contrasts) and key: separate file TableS1_contrast_grid.xlsx	
Figure S1	Database agreement score, study-cluster depth, invalid PMIDs and non-canonical residues	2
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Figure S1



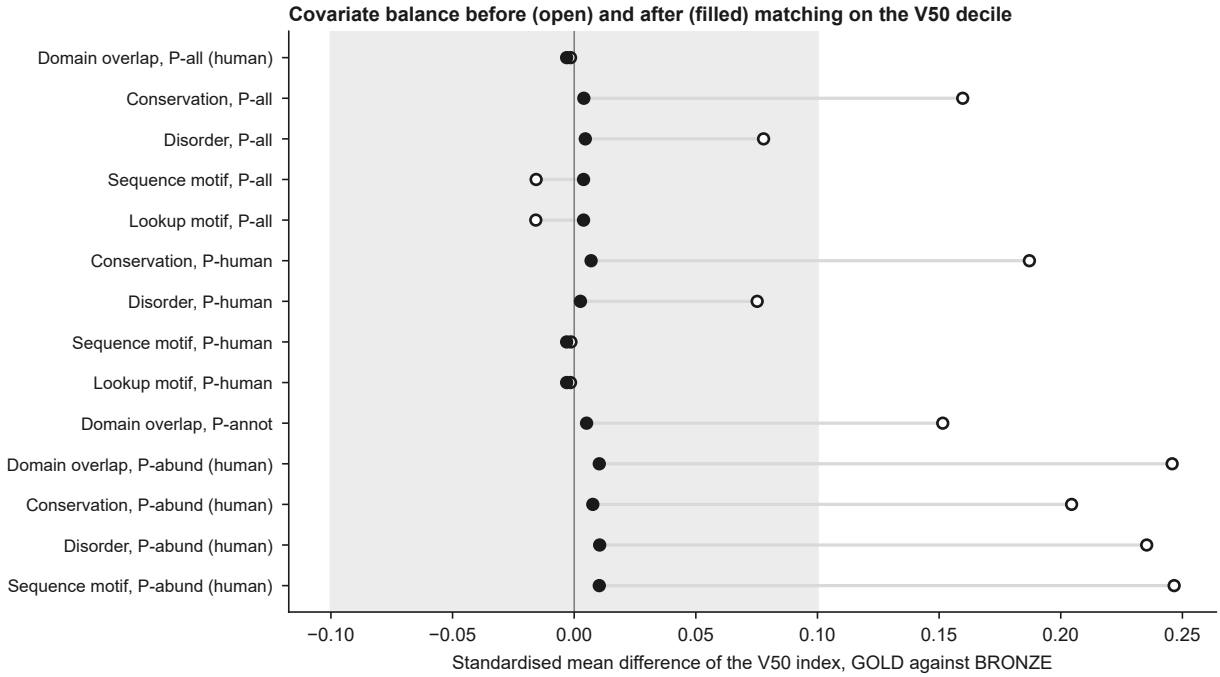
A: all 1,768,068 sites; source set B counts the PhosphoSitePlus, Phospho.ELM and dbPTM flags (score 0 to 3). B: sites with the study-cluster depth, and with the non-primary PMIDs not counted; dotted line: median depth. C: identifiers above 45,000,000 are treated as invalid; identifiers of valid format without a year were not found. D: identifiers above 45,000,000 are treated as invalid; identifiers of valid format without a year were not found. E: 46 sites in all.

Figure S2



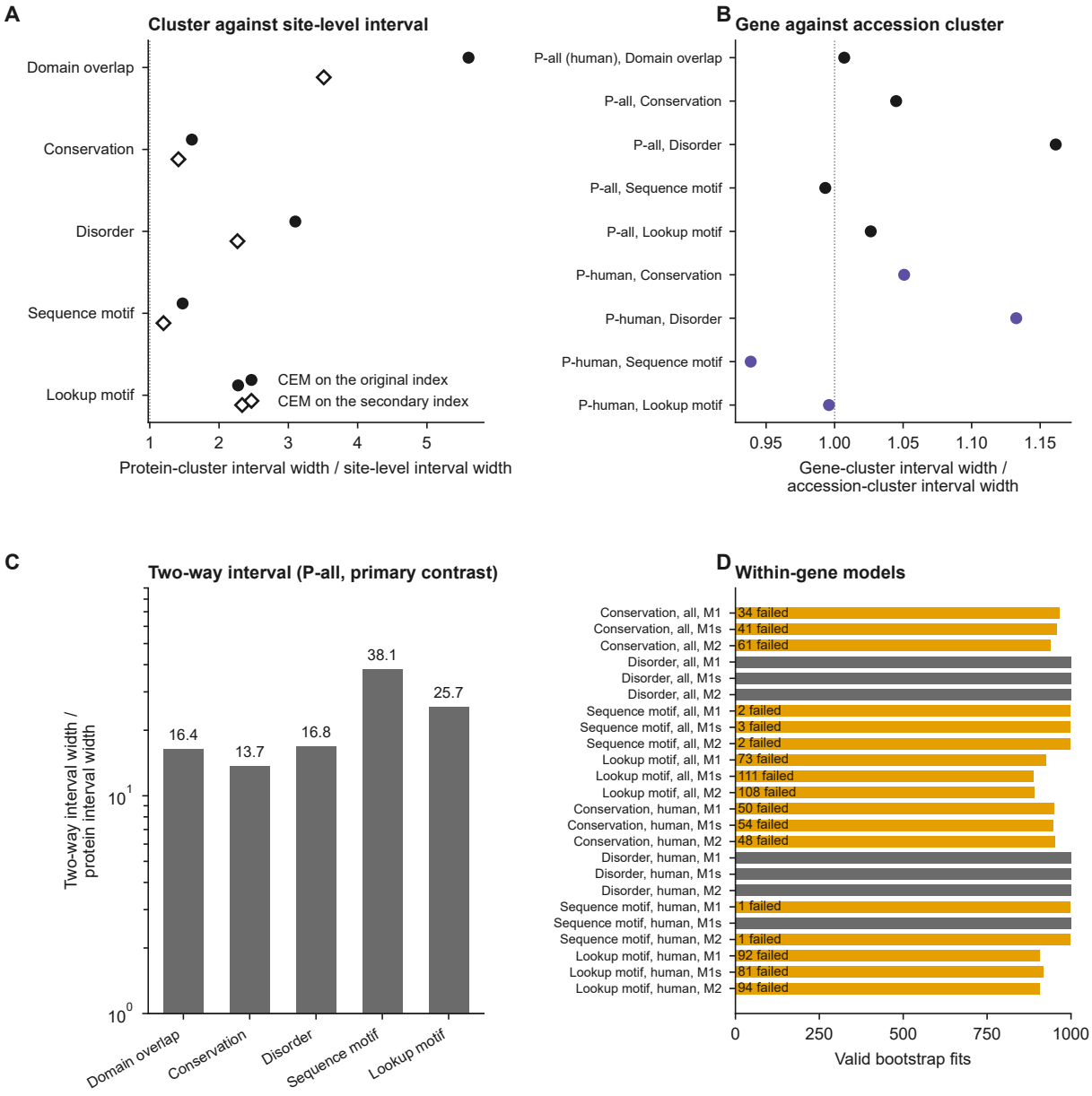
A: boxes span the interquartile range, whiskers the 5th to 95th percentile, open circles are means; human 48,162, mouse 28,918, rat 10,258 proteins. B: boxes span the interquartile range, whiskers the 5th to 95th percentile, open circles are means; proteins by highest tier: GOLD 34,883, SILVER 31,549, BRONZE 20,906; no interval is drawn on the means; the eta-squared interval is the stored protein-bootstrap interval. C: boxes span the interquartile range, whiskers the 5th to 95th percentile, open circles are means; proteins by highest tier: GOLD 34,883, SILVER 31,549, BRONZE 20,906; no interval is drawn on the means; the secondary index is built from the tiers' own publications, which is why it was not used for matching. D: counts per bin. E: 48,162 human proteins; annotation presence is a proxy for the outcome, not an independent variable. F: 48,162 human proteins, 18,757 gene clusters; Pharos target classes, V50 index. G: GOLD and BRONZE sites in P-all and P-human; 95% gene-cluster intervals. H: 48,162 human proteins, 18,757 gene clusters.

Figure S3



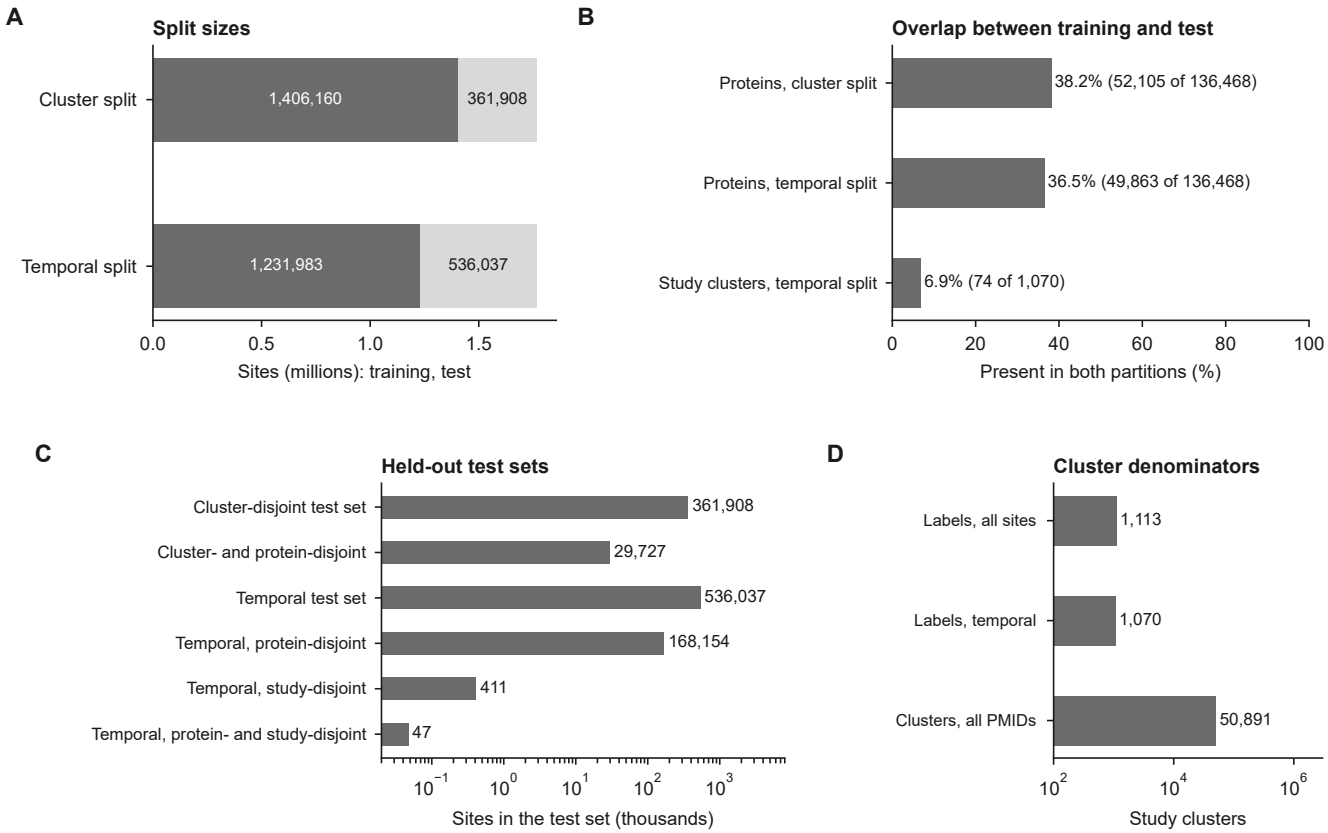
Shaded band: the balance criterion of the plan. CEM on the V50 decile, set-B tiers; the secondary-index design is not used.

Figure S4



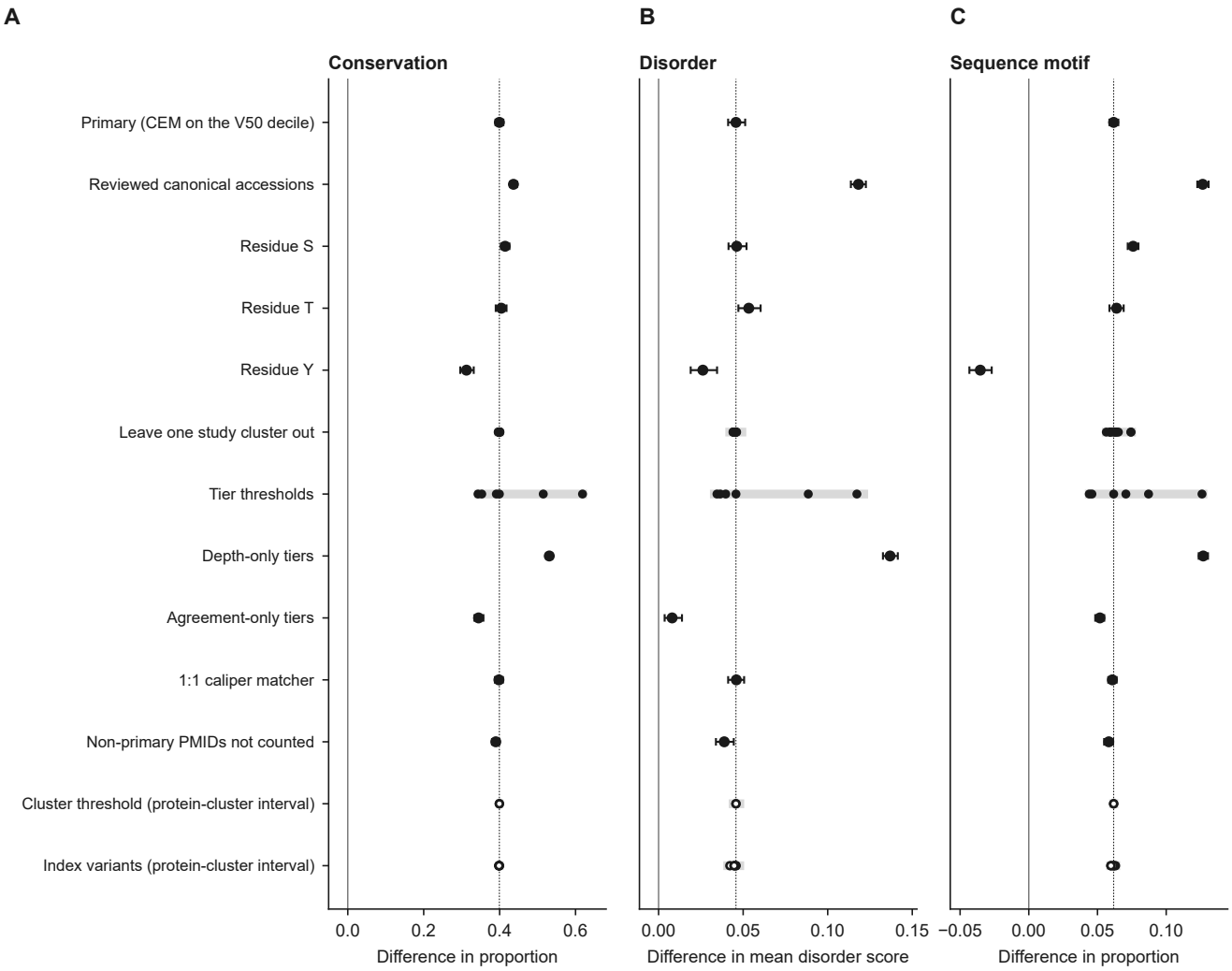
A: ratio of the protein-cluster interval to the site-level interval for the same estimate (estimates from the site-level analysis; compares interval methods only). B: gene-cluster against accession-cluster interval, D1 contrasts, populations P-all (black) and P-human (purple). C: ratio of the width of the two-way (protein x study) interval to the width of the protein-cluster interval, primary contrast in P-all. D: out of 1,000 resamples; orange bars mark rows with failed fits.

Figure S5



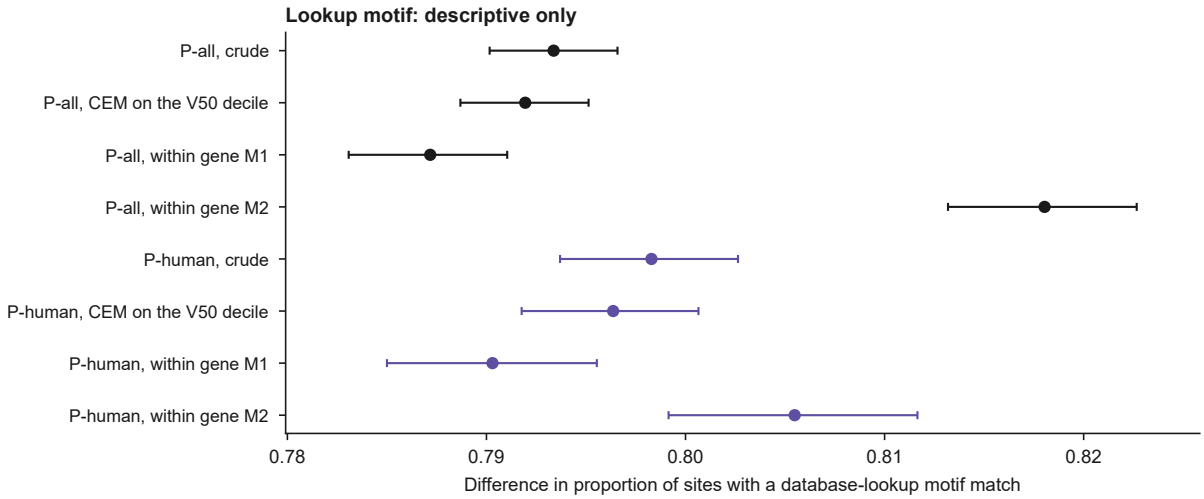
A: cluster split and temporal split of the 136,468 proteins. B: cluster split and temporal split of the 136,468 proteins. C: test sets as assigned before scoring. D: the number of study clusters depends on what is counted: distinct primary cluster labels (all sites, and sites in the temporal split) or clusters formed over all 50,904 distinct PubMed IDs under the merge rule.

Figure S6



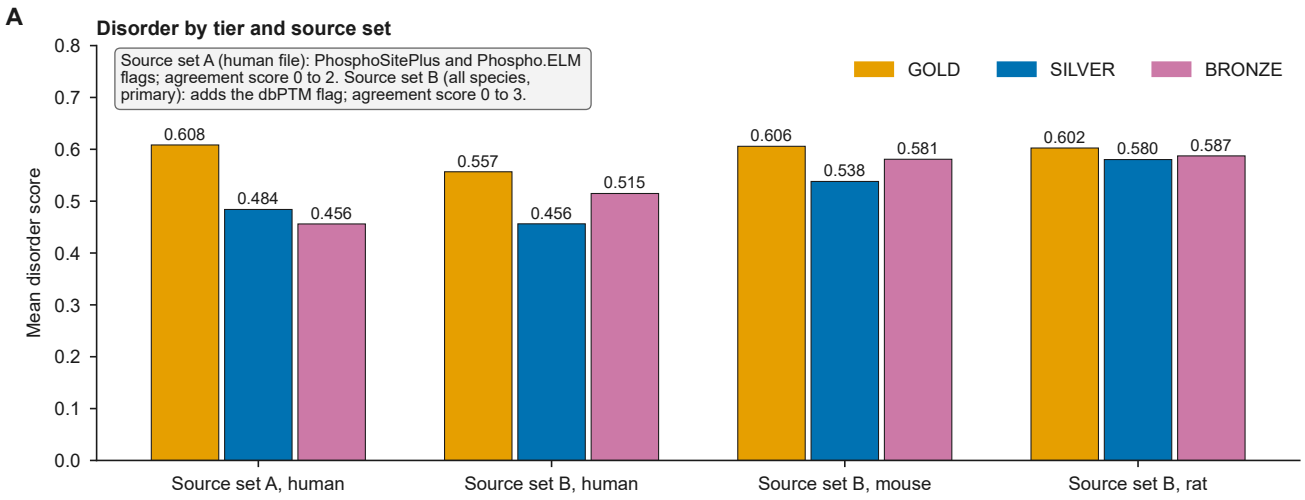
GOLD against BRONZE in P-all, set-B tiers unless named. Intervals are 95% gene-cluster intervals, except the last two rows (stored protein-cluster intervals, open circles; the cluster thresholds 500 and 1,000 sites change no record, so their estimates equal the primary estimate). Rows with several variants (leave one study cluster out, tier thresholds, cluster threshold, index variants) show every variant estimate as a dot and the range of the interval limits as a grey bar. Dotted vertical line: primary estimate.

Figure S7



Descriptive only, not graded: the indicator comes from the annotation tables that also feed the tiers (circular) and separates the tiers almost completely. GOLD against BRONZE, set-B tiers, 95% gene-cluster intervals; P-all: 324,907 / 478,880 sites, 37,903 gene clusters.

Figure S8



Means, no interval. Sites GOLD / SILVER / BRONZE: source set A, human 157,785 / 306,844 / 607,096; source set B, human 216,751 / 424,012 / 430,962; source set B, mouse 101,319 / 210,949 / 221,978; source set B, rat 24,369 / 115,841 / 21,887. Source set A: tiers from the human file with two source flags; source set B: three source flags.

Table S2. Main results for the human-only population (P-human)

Signal	Crude difference (95% CI)	Primary: CEM on the V50 decile (95% CI)	Within gene, M1 (95% CI)	Two-way interval excludes the null	GOLD / BRONZE sites; gene clusters	Grade
Conservation	0.4560 (0.4418 to 0.4691)	0.4501 (0.4349 to 0.4642)	0.4070 (0.3955 to 0.4181)	not computed	49,939 / 17,477; 7,965	moderate
Disorder	0.0542 (0.0468 to 0.0618)	0.0534 (0.0474 to 0.0611)	0.0473 (0.0443 to 0.0502)	not computed	208,322 / 129,438; 17,144	moderate
Sequence motif	0.0331 (0.0293 to 0.0370)	0.0368 (0.0331 to 0.0404)	0.0479 (0.0447 to 0.0511)	not computed	211,848 / 324,000; 17,640	moderate
Lookup motif (descriptive only)	0.7983 (0.7937 to 0.8026)	0.7964 (0.7918 to 0.8007)	0.7903 (0.7850 to 0.7955)	not computed	211,848 / 324,037; 17,641	descriptive only; not graded

Each estimate is GOLD minus BRONZE with set-B tiers: a difference in proportion (conservation, sequence motif, lookup motif) or in mean score (disorder), and an odds ratio for domain overlap (no difference is 0, and 1 for the odds ratio).

Intervals are 95% gene-cluster bootstrap intervals from 1,000 resamples. Crude: no adjustment, the same records as the primary column. Primary: coarsened exact matching (CEM) on deciles of the V50 study-effort index. Within gene, M1: GOLD against BRONZE within the same gene, all records of the population; not run for domain overlap.

Two-way interval excludes the null: whether the interval that allows for dependence among proteins and among studies (protein x study) excludes the null value; not computed for the human-only population. GOLD / BRONZE sites; gene clusters: records and gene clusters in the primary estimate.

Grade: strength of the claim under the grading rules of the claims table (strong, moderate, weak, not supported); the database-lookup motif is descriptive only because its indicator comes from the annotation tables that also feed the tiers, and is not graded.

Domain overlap is defined for human records only; its estimate is the one in Table 2.

Table S3. PMIDs that are not phosphorylation evidence: removed from the publication depth, or kept

PMID	Title	Records	Treatment	Reason or share
15174125	PhosphoSite: A bioinformatics resource dedicated to physiological protein phosphorylation.	219,383	removed from depth	title calls it a bioinformatics resource (PhosphoSite)
15489334	The status, quality, and expansion of the NIH full-length cDNA project: the Mammalian Gene Collection (MGC).	2,626	removed from depth	title is a project report on a gene collection (Mammalian Gene Collection), a resource
16381945	dbPTM: an information repository of protein post-translational modification.	82,357	removed from depth	title: an information repository (dbPTM)
18988627	Human Protein Reference Database--2009 update.	55,853	removed from depth	named database with an update marker (Human Protein Reference Database, 2009 update)
19060867	PhosphoPep--a database of protein phosphorylation sites in model organisms.	20,397	removed from depth	title: a database (PhosphoPep)
19366988	SysPTM: a systematic resource for proteomic research on post-translational modifications.	7,974	removed from depth	title: a systematic resource (SysPTM)
20428315	PhosphoGRID: a database of experimentally verified in vivo protein phosphorylation sites from the budding yeast Saccharomyces cerevisiae.	35,520	removed from depth	title: a database (PhosphoGRID)
21062810	Phospho.ELM: a database of phosphorylation sites--update 2011.	34,495	removed from depth	title: a database, update 2011 (Phospho.ELM)
21081558	PHOSIDA 2011: the posttranslational modification database.	49,545	removed from depth	title: the posttranslational modification database (PHOSIDA 2011)
21082442	Phosphoproteome resource for systems biology research.	52,111	removed from depth	title: a phosphoproteome resource
22135298	PhosphoSitePlus: a comprehensive resource for investigating the structure and function of experimentally determined post-translational modifications in man and mouse.	245,532	removed from depth	title: a comprehensive resource (PhosphoSitePlus)
23193284	PTMcode: a database of known and predicted functional associations between post-translational modifications in proteins.	127,502	removed from depth	title: a database (PTMcode)
23193290	DbPTM 3.0: an informative resource for investigating substrate site specificity and functional association of protein post-translational modifications.	56,670	removed from depth	named database with a release marker, title also says informative resource (DbPTM 3.0)
24174276	Comprehensive proteomic data sets for studying adipocyte-macrophage cell-cell communication.	5,343	removed from depth	title: proteomic data sets as the subject
24705204	SysPTM 2.0: an updated systematic resource for post-translational modification.	40,010	removed from depth	title: a systematic resource for modifications (SysPTM 2.0)
25514926	PhosphoSitePlus, 2014: mutations, PTMs and recalibrations.	225,187	removed from depth	named database with a year marker and update topics (PhosphoSitePlus, 2014: mutations, PTMs and recalibrations)
26425664	Dataset from the global phosphoproteomic mapping of early mitotic exit in human cells.	17,813	removed from depth	title: a dataset paper (Dataset from the global phosphoproteomic mapping ...)
29145615	iPTMnet: an integrated resource for protein post-translational modification network discovery.	421,351	removed from depth	title: an integrated resource (iPTMnet)
30087585	Residual tissue repositories as a resource for population-based cancer proteomic studies.	4,120	removed from depth	title: tissue repositories as a resource
30395287	UniProt: a worldwide hub of protein knowledge.	52,022	removed from depth	title: a hub of protein knowledge (UniProt), a knowledgebase
30418626	dbPTM in 2019: exploring disease association and cross-talk of post-translational modifications.	448,403	removed from depth	named database with a year marker (dbPTM in 2019)
30445427	15 years of PhosphoSitePlus®: integrating post-translationally modified sites, disease variants and isoforms.	219,383	removed from depth	named database with an anniversary marker (15 years of PhosphoSitePlus)
30476227	The BioGRID interaction database: 2019 update.	35,520	removed from depth	title: an interaction database, 2019 update (BioGRID)
32508104	Scop3P: A Comprehensive Resource of Human Phosphosites within Their Full Context.	93,664	removed from depth	title: a comprehensive resource (Scop3P)
32618424	LymphoAtlas: a dynamic and integrated phosphoproteomic resource of TCR signaling in primary T cells reveals ITSN2 as a regulator of effector functions.	5,532	removed from depth	title: a phosphoproteomic resource (LymphoAtlas)
35608653	In Vitro Kinase-to-Phosphosite Database (iKiP-DB) Predicts Kinase Activity in Phosphoproteomic Datasets.	12,725	removed from depth	title: a database (iKiP-DB)

37582339	Proteogenomic data and resources for pan-cancer analysis.	222,623	removed from depth	title: data and resources for pan-cancer analysis (also PubMed type Review, criterion a)
37619559	A proteogenomics data-driven knowledge base of human cancer.	222,623	removed from depth	title: a data-driven knowledge base of human cancer
30243273	An efficient pair natural orbital based configuration interaction scheme for the calculation of open-shell ionization potentials.	3,786	kept (not phosphorylation evidence, rule unchanged)	0.21% of records
37031204	Correction to: Phospho-heavy-labeled-spiketide FAIMS stepped-CV DDA (pHASED) provides real-time phosphoproteomics data to aid in cancer drug selection.	3,893	kept (not phosphorylation evidence, rule unchanged)	0.22% of records
14702039	Complete sequencing and characterization of 21,243 full-length human cDNAs.	1,864	kept (not phosphorylation evidence, rule unchanged)	0.11% of records

Records: site records whose publication list contains the PMID. Treatment: removed from depth = the PMID is not counted in study-cluster depth or in the tiers (sensitivity analysis); kept = the PMID is not phosphorylation evidence but stays in the count, because the rule is unchanged.

Reason or share: the reason a PMID was classed as a database, resource or review publication, or the share of all 1,768,068 records for a kept PMID. Titles are quoted from PubMed.

Table S4. UniProt experimental against inferred modified-residue annotations

Signal	Design	Experimental minus inferred (95% gene-cluster CI)	Experimental / inferred records; gene clusters
Domain overlap	crude	1.336 (1.138 to 1.583)	5,021 / 9,106; 4,522
Domain overlap	CEM on the V50 decile	1.026 (0.874 to 1.229)	5,021 / 9,106; 4,522
Domain overlap	index x abundance deciles	1.068 (0.887 to 1.243)	5,003 / 9,048; 4,492
Conservation	crude	-0.1757 (-0.2166 to -0.1412)	1,467 / 2,586; 1,698
Conservation	CEM on the V50 decile	-0.1297 (-0.1881 to -0.0873)	1,467 / 2,586; 1,698
Conservation	index x abundance deciles	-0.1561 (-0.2014 to -0.1098)	1,463 / 2,576; 1,688
Disorder	crude	-0.0637 (-0.0797 to -0.0483)	5,014 / 9,091; 4,513
Disorder	CEM on the V50 decile	-0.0318 (-0.0489 to -0.0147)	5,014 / 9,091; 4,513
Disorder	index x abundance deciles	-0.0346 (-0.0465 to -0.0123)	4,999 / 9,035; 4,484
Sequence motif	crude	-0.0217 (-0.0446 to -0.00004)	5,021 / 9,106; 4,522
Sequence motif	CEM on the V50 decile	0.0327 (0.0087 to 0.0545)	5,021 / 9,106; 4,522
Sequence motif	index x abundance deciles	0.0383 (0.0134 to 0.0674)	5,003 / 9,048; 4,492

Experimental and inferred refer to the evidence code (ECO) of the UniProt modified-residue annotation of the site. Estimates are experimental minus inferred, as a difference in proportion or mean score, and as an odds ratio for domain overlap; intervals are 95% gene-cluster bootstrap intervals.

A limit that would print as zero at the usual number of decimals is given with more decimals so that its sign shows.

Table S5. Sample sizes and estimates behind Figure 4A

Population	Signal	Tier definition	GOLD sites	BRONZE sites	Gene clusters	Crude (95% CI)	CEM on the V50 decile (95% CI)
P-human	Domain overlap	set-B tiers	211,848	324,037	17,641	6.065 (5.731 to 6.387)	5.912 (5.588 to 6.222)
P-human	Domain overlap	set-A tiers	153,738	483,427	18,434	2.494 (2.346 to 2.644)	2.646 (2.491 to 2.803)
P-human	Domain overlap	depth-only tiers	153,571	483,453	18,434	2.492 (2.343 to 2.641)	2.644 (2.487 to 2.799)
P-human	Domain overlap	agreement-only	174,814	412,834	17,698	8.099 (7.620 to 8.546)	7.581 (7.122 to 8.008)
P-human	Conservation	set-B tiers	49,939	17,477	7,965	0.4560 (0.4422 to 0.4692)	0.4501 (0.4360 to 0.4645)
P-human	Conservation	set-A tiers	35,760	49,705	8,700	0.6241 (0.6142 to 0.6333)	0.6162 (0.6064 to 0.6255)
P-human	Conservation	depth-only tiers	35,718	49,709	8,700	0.6245 (0.6152 to 0.6331)	0.6165 (0.6066 to 0.6257)
P-human	Conservation	agreement-only	43,671	19,757	8,018	0.3978 (0.3803 to 0.4142)	0.3920 (0.3756 to 0.4090)
P-human	Disorder	set-B tiers	208,322	129,438	17,144	0.0542 (0.0466 to 0.0610)	0.0534 (0.0466 to 0.0606)
P-human	Disorder	set-A tiers	150,249	288,135	18,030	0.1614 (0.1557 to 0.1670)	0.1507 (0.1451 to 0.1566)
P-human	Disorder	depth-only tiers	150,082	288,149	18,030	0.1616 (0.1555 to 0.1678)	0.1508 (0.1448 to 0.1570)
P-human	Disorder	agreement-only	174,728	164,082	17,201	0.0052 (-0.0034 to 0.0141)	0.0107 (0.0044 to 0.0191)
P-human	Sequence motif	set-B tiers	211,848	324,000	17,640	0.0331 (0.0295 to 0.0368)	0.0368 (0.0334 to 0.0403)
P-human	Sequence motif	set-A tiers	153,738	483,390	18,434	0.1035 (0.0999 to 0.1073)	0.1008 (0.0973 to 0.1045)
P-human	Sequence motif	depth-only tiers	153,571	483,416	18,434	0.1035 (0.1000 to 0.1073)	0.1008 (0.0974 to 0.1047)
P-human	Sequence motif	agreement-only	174,814	412,796	17,697	0.0177 (0.0139 to 0.0215)	0.0264 (0.0231 to 0.0299)
P-all	Conservation	set-B tiers	93,308	37,591	17,777	0.3986 (0.3884 to 0.4094)	0.3993 (0.3897 to 0.4097)
P-all	Conservation	depth-only tiers	66,155	113,047	21,077	0.4734 (0.4654 to 0.4815)	0.5308 (0.5229 to 0.5385)
P-all	Conservation	agreement-only	81,550	43,466	17,851	0.3348 (0.3213 to 0.3486)	0.3445 (0.3336 to 0.3573)
P-all	Disorder	set-B tiers	320,780	215,745	35,524	0.0438 (0.0384 to 0.0493)	0.0457 (0.0412 to 0.0514)
P-all	Disorder	depth-only tiers	229,448	506,153	41,199	0.1393 (0.1351 to 0.1438)	0.1369 (0.1327 to 0.1415)
P-all	Disorder	agreement-only	270,355	270,815	35,699	0.0018 (-0.0039 to 0.0070)	0.0080 (0.0036 to 0.0138)
P-all	Sequence motif	set-B tiers	324,907	478,837	37,901	0.0617 (0.0583 to 0.0652)	0.0618 (0.0585 to 0.0650)
P-all	Sequence motif	depth-only tiers	233,219	771,537	43,391	0.1314 (0.1281 to 0.1349)	0.1270 (0.1237 to 0.1304)
P-all	Sequence motif	agreement-only	270,840	604,860	38,103	0.0469 (0.0437 to 0.0505)	0.0518 (0.0484 to 0.0549)

The cells behind Figure 4A: GOLD against BRONZE under each tier definition, with the numbers of sites and gene clusters. Intervals are 95% gene-cluster bootstrap intervals; domain overlap is an odds ratio, the other signals are differences.